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Review Article

# Corrosion behavior of magnesium metal matrix in different corrosive media – A review

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## Article

## Abstract

Magnesium metal matrix composites (Mg-MMCs) are emerging as promising lightweight structural materials for automotive, aerospace, and biomedical applications because of their high specific strength, low density, and improved mechanical performance compared with monolithic Mg alloys. However, their use is limited by susceptibility to corrosion in aggressive environments, particularly in chloride-rich and marine environments. In this review, critical gaps and unresolved challenges that continue to hinder the technological deployment of Mg-MMCs were identified and addressed, including the scarcity of long-term corrosion data under realistic service environments, limited understanding of reinforcement–matrix interfacial evolution during corrosion, and the persistent trade-off between mechanical strengthening and corrosion resistance.

## Outline

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Mg-MMC corrosion fundamentals were highlighted, discussing various corrosion mechanisms, such as galvanic corrosion, pitting corrosion, filiform corrosion, and crevice corrosion, in various aggressive media (acidic, alkaline, saline (marine), industrial, and biological media). Furthermore, corrosion prevention techniques were identified and highlighted, including surface treatments, alloying strategies, and reinforcement modifications for corrosion protection. Among the mentioned surface treatments, plasma electrolytic oxidation and sol-gel coatings generally exhibit the highest compatibility because they can effectively insulate the conductive reinforcement-matrix interface. The microstructural behavior of Mg-MMCs in corrosive environments acts as a fingerprint for the underlying electrochemical mechanism of the composites. The review concludes with research perspectives aimed at enhancing corrosion resistance and ensuring broader adoption of Mg-MMCs in advanced engineering systems.

Keywords

Magnesium metal matrix composites; Corrosion behavior; Saline environment; Acidic medium; Surface modification; Protective coatings

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1. Introduction

Magnesium (Mg), recognized as the lightest structural metal, with a density of 1.74 g/cm<sup>3</sup> [1], possesses remarkable attributes, including high specific strength of about 130 kNm/kg [2], excellent damping capacity [3], and good machinability [4]. Magnesium is advantageous because of its bioactivity, biodegradability, biocompatibility, and its resemblance to the elastic modulus of natural bone. The low elastic modulus of Mg-based materials aids in reducing the stress shielding effect, which majorly affects stainless steel, cobalt-chromium alloy, and titanium-based implant material [5]. The lightweight and low

7. Conclusion

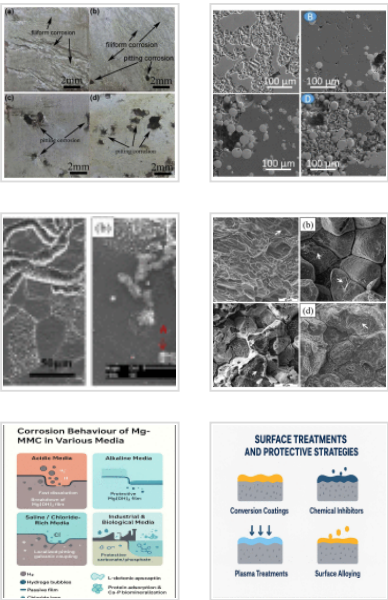
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density of magnesium make it possess excellent mechanical strength and fracture toughness, with significantly poor corrosion resistance. Therefore, the high susceptibility to oxidation and low corrosion resistance restricted the use of magnesium and its alloys [5]. Magnesium is naturally non-toxic when used in biomedical applications and does not produce any inflammation. Magnesium possesses numerous advantages, applications, and shortcomings that are highlighted in various studies [5,6]. Magnesium alloys are used as biodegradable implants providing bone remodeling, tissue healing rate, and injured bone tabulating. They are employed in bone and cardiovascular applications for fracture restoration. Besides Mg, Titanium and its alloys have been broadly utilized in biomedical research and applications since the mid-20th century because of their high specific strength, low density, excellent corrosion resistance, and outstanding biocompatibility [7]. However, its limited strength and creep resistance have necessitated the development of magnesium metal matrix composites (Mg-MMCs), where Mg is reinforced with ceramic particles, fibers, or carbon-based materials [8]. The reinforcement addition in the Mg matrix could be monolithic or hybrid, leading to Mg-MMCs development. The inclusion of reinforcement to Mg results in enhanced mechanical behaviour with high wear and corrosion resistance. This leads to beneficial aspects of MMCs, such as high thermal stability, good tribological properties, superior electrical and damping properties, and controllable thermal expansion coefficient [5]. These composites demonstrate superior mechanical performance, such as high stiffness and high specific strength [9], enhanced wear resistance, and improved thermal stability compared to conventional Mg alloys [10]. Owing to their low density, favorable strength-to-weight ratio, biocompatibility, and outstanding electromagnetic shielding properties, Mg-MMCs are widely applied in industries such as automotive, aerospace, defense, and marine engineering [11,12]. Nevertheless, corrosion continues to be a critical challenge restricting their long-term usage and broader adoption [13]. The high reactivity of Mg, combined with galvanic interactions between the matrix and reinforcement phases, accelerates localized corrosion in aggressive environments [14]. Recent investigations have shown that reinforcement type, particle size distribution, and interfacial integrity play decisive roles in corrosion kinetics and degradation mechanisms [15,16]. Furthermore, surface-

engineered Mg-based composites such as plasma-electrolytic-oxidized (PEO) and laser-treated Mg-MMCs have attracted growing research interest due to their demonstrated ability to mitigate corrosion through barrier protection, improved adhesion, and controlled electrochemical behavior [17]. In response to these challenges, considerable progress has been made in understanding corrosion mechanisms and developing protection strategies, including alloying, surface modification, and hybrid coatings [18,19].

A study by [20] investigated the use of friction stir processing (FSP) as a post-treatment method to enhance the corrosion resistance of low-pure cold sprayed (LPCS) Al-Al<sub>2</sub>O<sub>3</sub> metal matrix composite coatings deposited on AZ31B magnesium alloy. Various analyses were carried out on the cold-sprayed and FSPed samples, including structural integrity, phase analysis, and corrosion studies. It was revealed that FSP improved the structural integrity of MMC coating, while FSP-treated samples showed superior corrosion resistance to cold-sprayed samples in 3.5 wt.% NaCl solution through the potentiodynamic polarization (PDP) and electrochemical impedance spectroscopy (EIS) tests. This enhancement was linked to the increase in the interfacial bonding between the matrix and reinforcement phase. In a related study, the enhancement observed in the FSP-treated samples was linked to alumina particle refinement/redistribution, reduced inter-splat boundaries, surface re-oxidation of aluminum, and decreased dislocation density in the matrix phase due to dynamic recrystallization. However, the corrosion resistance deteriorated due to the excessive processing, which damaged the interior coating interfaces and increased dislocation density without further beneficial particle breakup [20]. To overcome the stress shielding effect and non-degradability commonly observed in metallic implants, a novel approach was introduced through hybrid metal/ceramic reinforcement within Mg-3Zn binary alloy via squeeze casting fabrication process [21]. The study incorporated 1 wt.% Titanium (Ti) and varying weight percentages (0.5 – 1.5 wt.%) of hydroxyapatite (HA) as reinforcement in the production of the MMC. Of interest is the corrosion performance of the hybrid reinforced Mg-MMC (Mg-3Zn/1Ti/1.5 HA) in a phosphate-buffered saline solution, showing comparable performance with the unreinforced alloy, with a linear polarization resistance of 893 ohms and a corrosion

density of  $4.53 \times 10^{-5} \mu\text{A}/\text{cm}^2$ . This hybrid reinforced MMC was recommended to be a promising material for load-bearing biodegradable implants.

The corrosion properties of stellite coated on AZ91D alloy by plasma spray technique were investigated by [22]. The corrosion immersion experiment was conducted using sodium chloride at different pH levels, chloride ion concentrations, and exposure times. The study revealed a significant influence of the pH value on the corrosion behaviour of the stellite coatings, followed by other parameters. The stellite coatings could not provide sufficient corrosion protection to Mg alloy during longer exposures in the corrosive medium. Stellite coatings were reported to be highly susceptible to localized degradation and could not provide adequate corrosion resistance to Mg alloy substrates in solutions containing higher chloride concentrations. The corrosion and cytotoxicity of biodegradable Mg-3Zn alloy reinforced with titanium and HA were evaluated by [23] for its orthopedic implant applications. Mg-MMC (Mg-3Zn/1Ti/1.5HA) showed 91.7% viability, indicating superior biocompatibility for orthopedic applications. The corrosion resistance of the composite was enhanced, showing minimal mass loss. Mg-3Zn/1Ti/1.5HA revealed a lower corrosion rate of 10% and 21% compared to Mg-3Zn/1Ti/1HA and Mg-3Zn/1Ti/0.5HA, respectively. This implies that the increase in the hydroxyapatite volume weight in the reinforcement has a significant impact on the corrosion of the composite. As reported by [23], the corrosion resistance of hybrid composites was increased with the inclusion of 1.5% hydroxyapatite. The corrosion resistance was enhanced owing to the refined grain structure and protective oxide layers formed by the Ti and HA incorporation into the matrix alloy.

Different methods have recently been used in producing novel Mg-MMCs to have better mechanical and corrosion properties. For biomedical applications, Mg-MMCs reinforced with graphene oxide (GO) pose a significant challenge due to the inherent trade-off between mechanical strength and corrosion resistance. As a result of this, [24] developed a novel hybrid technique based on pre-compositing interfacial modification and ultrasonically modulated process. The study applied polyvinyl alcohol coating on the ZM5 Mg alloy substrates, improving their wettability and interfacial compatibility with GO. It

was reported that there is a strategic distribution of the GO sheets together with precipitates along grain boundaries. This results in continuous barrier formation that effectively inhibits corrosive media penetration while suppressing galvanic corrosion. In the study of [25], the thixomolding process was used in the development of Mg-MMC with graphene nanoplatelets (GNPs) as reinforcement. The thixomolding process is an extremely efficient dispersing technique, which was used to distribute the GNP reinforcements at 1, 0.1, 0.3, 0.6, 0.9, and 1.2 wt.% in the AZ91D magnesium alloy. The effects of some factors, such as screw rotation rate and injection velocity on microstructures, GNPs dispersibility, and mechanical properties were examined. However, corrosion studies were not investigated. The study revealed the establishment of strong bonding interfaces between the matrix and the GNPs reinforcement due to the formed MgO nanoparticles that form Mg/MgO semi-coherency interface and MgO/GNPs interface. The developed technology has potential for large-scale production of both thin- and thick-walled complex products. The thixomolding process was employed to fabricate Mg-MMCs using GNPs. However, the microstructure evolution of reinforcement was investigated in the study [25]. Through the fabrication technique adopted, the mechanical properties of the composites were enhanced. The interfacial load transfer was improved due to the increased load-bearing capability and the effective interfacial shear strength. There is an intergranular reinforcement by finer particles, which could bring positive and directive significance for the structural design of the composites. In a similar study by [26], the corrosion study of the thixomolding-processed Mg-MMCs was investigated. It was reported that the galvanic corrosion of the matrix was inhibited while the corrosion resistance of the composites improved. The unique distribution of the GNPs along grain boundaries plays a vital role in improving the corrosion resistance. The outcome of the study broadened the horizon for biomedical applications. The serpentine channel pouring (SCP) process is another novel method that could be used to produce Mg-MMCs with enhanced properties and has wide industrial applications [27]. The introduction and utilization of these fabrication technologies have facilitated the production of Mg-MMCs with better mechanical properties and improved corrosion resistance.

Although several review articles have addressed corrosion mechanisms and protection strategies for Mg-MMCs, most existing reviews offer fragmented analyses, often focusing narrowly on either reinforcement effects or surface treatments while overlooking their interdependence [28]. Consequently, a comprehensive, up-to-date synthesis that integrates bulk composite design [29], interfacial phenomena [30], and surface modification within a unified corrosion framework remains lacking [31]. This review offers distinct insights by systematically correlating reinforcement characteristics [32], interfacial chemistry [33,34], and electrochemical response with corrosion behavior in Mg-MMCs [35]. Unlike prior studies that treat surface engineering as an empirical remedy, this review critically evaluates modern protection strategies, such as PEO-based duplex coatings [[36], [37], [38]].

Importantly, this review identifies critical gaps and unresolved challenges that continue to hinder the technological deployment of Mg-MMCs. These include the scarcity of long-term corrosion data under realistic service conditions, a limited understanding of the reinforcement–matrix interfacial evolution during corrosion, and the persistent trade-off between mechanical strengthening and corrosion resistance. Furthermore, the lack of standardized corrosion testing methodologies complicates cross-comparison of reported results, underscoring the need for unified evaluation frameworks.

Therefore, a comprehensive understanding of their corrosion mechanisms in various media is vital to improving design and developing effective protection strategies. This paper highlights recent progress in the study of Mg-MMC corrosion behavior across different environments and examines protective measures aimed at enhancing their durability. The novelty of this review is its systematic integration of three critical dimensions of reinforcement characteristics, interfacial chemistry, and electrochemical response into a unified corrosion framework for magnesium metal matrix composites (Mg-MMCs). Unlike previous studies that offered fragmented analyses, this work establishes comprehensive correlations between interfacial phenomena and surface modification strategies. Specifically, it critically evaluates modern protection approaches

such as plasma electrolytic oxidation (PEO)-based duplex coatings, moving beyond empirical surface treatment remedies to provide a mechanistic understanding.

## 2. Fundamentals of Mg-MMC and corrosion

Magnesium alloys typically contain elements such as aluminium (Al), zinc (Zn), manganese (Mn), silicon (Si), zirconium (Zr), copper (Cu), and rare earth metals [39]. A wide range of alloys is available, including casting alloys such as AZ63, AZ81, AZ91, AM50, AM60, ZK51, ZK61, ZE41, and ZC63, as well as wrought alloys like AZ31, AZ61, AZ80, ZK60, ZE41, and ZC71. These alloys serve as the primary base materials for producing magnesium metal matrix composites (Mg-MMCs). The major alloying elements in magnesium alloys are summarized in Table 1, each contributing specific benefits such as improvements in strength, hardness, ductility, casting fluidity, wear resistance, and corrosion resistance.

Table 1. Main Alloying elements with ASTM [designation ↗](#) of magnesium alloys [40].

| S/N | Alloying Element                | ASTM <a href="#">Designation ↗</a> |
|-----|---------------------------------|------------------------------------|
| 1   | Aluminum-Manganese              | <a href="#">AM ↗</a>               |
| 2   | Aluminum-Zinc-Manganese         | <a href="#">AZ ↗</a>               |
| 3   | Aluminum-Silicon-Manganese      | <a href="#">AS ↗</a>               |
| 4   | Rare Earth Metal-Zirconium      | <a href="#">EZ ↗</a>               |
| 5   | Zinc-Copper-Manganese           | <a href="#">ZC ↗</a>               |
| 6   | Zinc-Zirconium-Rare Earth Metal | <a href="#">ZE ↗</a>               |
| 7   | Zinc-Zirconium                  | <a href="#">ZK ↗</a>               |
| 8   | Manganese                       | <a href="#">M ↗</a>                |

| S/N | Alloying Element | ASTM Designation ↗ |
|-----|------------------|--------------------|
| 9   | Zirconium        | K ↗                |

The alloying elements in magnesium serve distinct functions, being incorporated to enhance properties. To further improve performance, various reinforcements can be introduced into magnesium alloys for the fabrication of Mg-MMCs. Common reinforcements include Titanium Carbide (TiC), Titanium Dioxide (TiO<sub>2</sub>), Tantalum Carbide (TaC), Titanium Nitride (TiN), Titanium Diboride (TiB<sub>2</sub>), Aluminium Nitride (AlN), Silicon Carbide (SiC), Silica (SiO<sub>2</sub>), Zirconium Carbide (ZrC), Zirconia (ZrO<sub>2</sub>), Zirconium Diboride (ZrB<sub>2</sub>), Zinc Oxide (ZnO), Alumina (Al<sub>2</sub>O<sub>3</sub>), Magnesium Oxide (MgO), Silicon Nitride (Si<sub>3</sub>N<sub>4</sub>), Calcium Hexaboride (CaB<sub>6</sub>), Tungsten Carbide (WC), Boron Carbide (B<sub>4</sub>C), Boron Nitride (BN), and graphite (C), with several of these materials frequently employed in past studies [41].

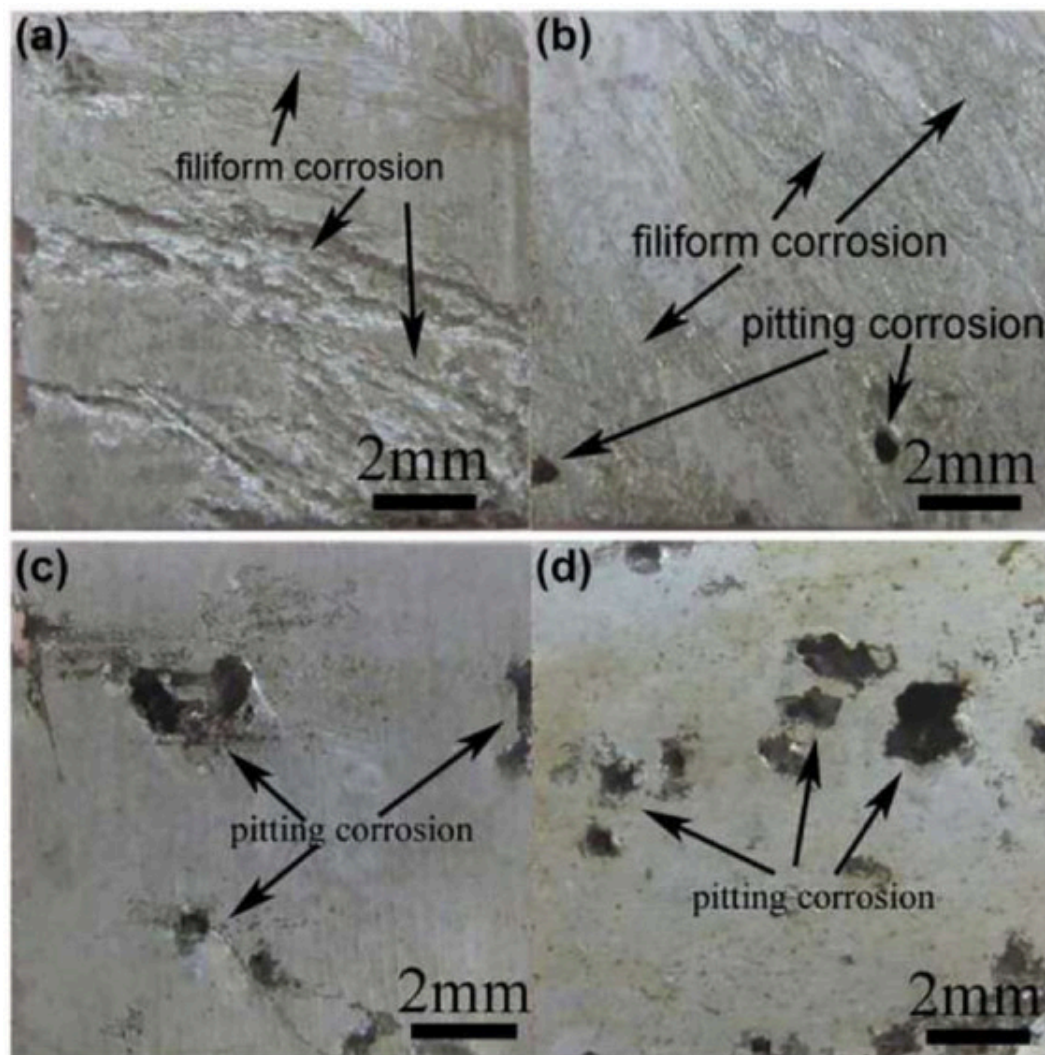
The corrosion behavior of Mg-MMCs is primarily controlled by galvanic coupling, as electrochemical potential differences between the magnesium matrix and the reinforcement particles promote micro-galvanic corrosion [42]. In chloride-rich environments, such as physiological fluids, magnesium undergoes rapid degradation characterized by localized pitting and hydrogen gas evolution, which can hinder healing processes and trigger adverse biological responses [43]. Multiple factors, including alloy composition, reinforcement type, interfacial bonding quality, and fabrication-related defects, determine the extent and severity of corrosion [44,45].

Some important parameters have been used to evaluate reinforcement from the perspective of both mechanical properties and corrosion resistance, which include morphology, volume fraction, size, distribution, orientation, and reactivity of the reinforcement. The Mg-MMCs are often classified according to the aspect ratio (length: diameter) of the reinforcement, as this indicates whether the reinforcement is particles, whiskers, or short fibers, continuous fibers, laminates, or sheets [44,46]. Recently, reinforcing particles, such as fluorapatite [47], tri-calcium phosphate [1], poly(lactic-co-

glycolic acid) [48], or hydroxyapatite [49], have been used to develop bio-degradable Mg-MMCs. These reinforcements serve to alter the corrosion rate of the Mg alloy and could also promote the formation of apatite layers on the Mg-MMCs surface during the bone-healing process. The incorporation of magnesium oxide (MgO) graphene nanosheets into magnesium as reinforcement in the development of Mg-MMCs led to decreased opening defects, micro-cracks, and micro-pores of the composites, which hinder the penetration of the corrosive solution into the matrix. Hence, the corrosion rate of the developed composites reduced with the introduction of the reinforcement particles [50].

Similar to other metals, the corrosion behavior of magnesium alloys can be broadly classified into uniform corrosion and localized corrosion when considered from electrochemical, compositional, and microstructural viewpoints at the macroscale. Localized corrosion primarily encompasses galvanic corrosion, pitting, filiform corrosion, intergranular corrosion (IGC), exfoliation corrosion (EFC), crevice corrosion, stress corrosion cracking (SCC), corrosion fatigue (CF), and erosion corrosion. However, exfoliation and crevice corrosion have been reported only infrequently.

Pitting corrosion is among the most prevalent forms of degradation in magnesium alloys. Pit initiation is closely associated with the alloy's chemical composition and particular microstructural characteristics present on the exposed surface. Factors such as composition, grain size, heat-treatment state, and the morphology and distribution of secondary phases all play significant roles in governing the corrosion behavior of Mg alloys. In sand-cast ZE41-T5 magnesium alloy exposed to an aggressive environment, pit initiation occurs within the  $\alpha$ -Mg matrix adjacent to the T-phase ( $\text{Mg}_7\text{Zn}_3\text{RE}$ ) along grain boundaries [51]. This behavior is mainly due to galvanic coupling arising from the potential difference between the T-phase and the  $\alpha$ -Mg matrix. With continued immersion, Zr-enriched regions at the centers of the grains become preferential sites for corrosion. Minor pits first form in these areas and subsequently evolve into more severe localized attacks. A microstructural examination of a typical Mg alloys indicating pitting corrosion, is displayed in Fig. 1.



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Fig. 1. Micrograph of filiform and pitting corrosions of a Mg-Dy alloy after immersion in 0.9 wt.% NaCl solution [51,52].

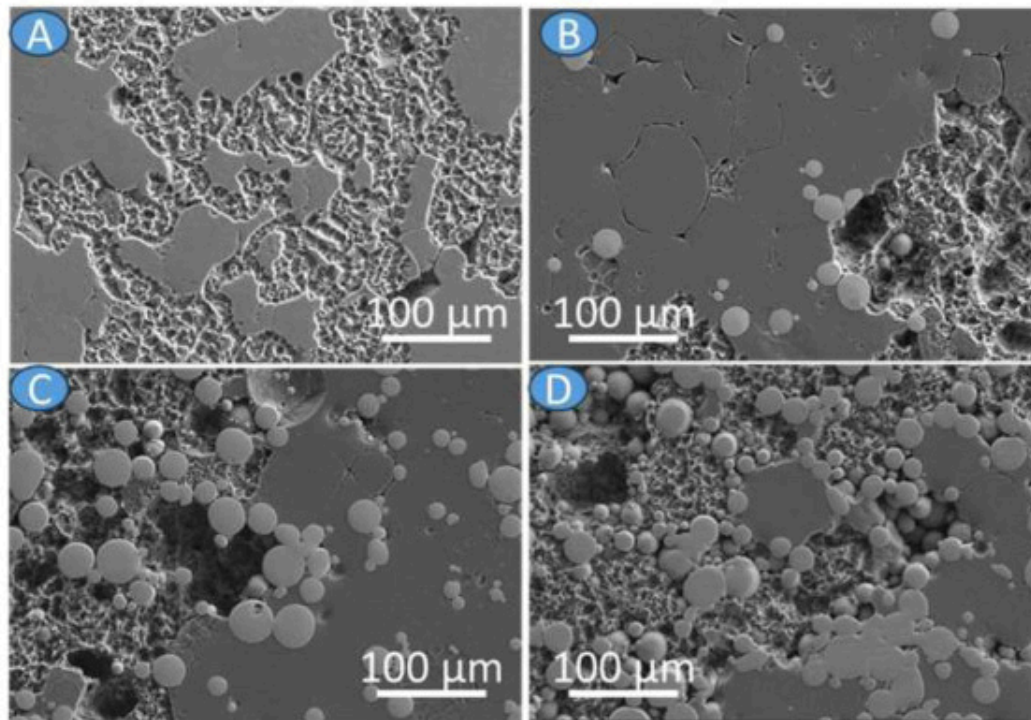
The pitting corrosion of Mg alloys is significantly linked to their microstructure, especially second phases. The second phase in traditional Mg alloys, such as AZ91, is

electrochemically nobler than Mg matrix, while the second phase in rare earth (RE) Mg alloy GW93 is more active than Mg matrix. As a result, the pitting corrosion mechanism of Mg alloy GW93 is different from the traditional ones [53]. studied the pitting corrosion mechanism of Mg alloy GW93 through various scientific means. Based on the results, the pitting corrosion of GW93 involves three stages: second-phase dissolution, then corrosion of the Mg matrix adjacent to the dissolved second phases, and finally, propagation of corrosion pits along the depth direction of the dissolved second phases. It was revealed that the anodic second phases and enrichment of  $\text{Cl}^-$  in the thick corrosion product films dominate the propagation of pitting corrosion. Pitting corrosion is influenced by grain size. When the grains are refined, the corrosion morphology changes from pitting to uniform corrosion.

Typically, the pitting process in magnesium alloys is fundamentally electrochemical, and it involves the breaking down of the protective oxide/hydroxide film with the establishment of active-passive cells. A typical electrochemical process within a corrosion pit in Mg-Al alloys reveals a separation of anodic and cathodic reactions, with metal dissolution occurring at the pit base and cathodic oxygen reduction on the surrounding passive surface (neutral/alkaline conditions). The autocatalytic nature is maintained by chloride ion migration toward the anodic site and hydrolysis reactions generating  $\text{H}^+$  ions.

The hydrolysis of dissolved metal ions produces hydrogen ions, creating an acidic microenvironment (pH 2 - 4) within the pit that prevents repassivation and accelerates dissolution. This autocatalytic mechanism distinguishes pitting from other corrosion forms. Pitting corrosion progresses through distinct morphological stages that can be quantified and modeled. Stage I is the initiation stage (nucleation), where there is a breakdown of  $\text{Mg}(\text{OH})_2/\text{MgO}$  passive film at weak points. The initiation sites could be at the intermetallic particles ( $\text{Mg}_{17}\text{Al}_{12}$ , Al-Mn phases), surface scratches and machining marks, grain boundaries, and reinforcement particles in Mg-MMCs. Stage II refers to the growth or propagation of the pits, and stage III is coalescence and penetration.

Galvanic corrosion is a prevalent corrosion mechanism in magnesium alloys. For instance, it can occur when magnesium components are joined or welded to steel parts in lightweight powertrain systems and automotive body structures. In certain situations, galvanic corrosion can develop between layers of the same alloy that differ in grain size. For example, hot-extruded ZK60 magnesium alloy exhibited coarse grains near the surface and finer grains in the interior [54]. The outer layer, which has a higher open-circuit potential (OCP), tends to become passivated, whereas the inner layer with a lower OCP undergoes preferential corrosion. Fig. 2 presents a typical galvanic corrosion phenomenon on AZ31-Ti6Al4V composites. This occurs as a result of strong micro-galvanic corrosion between AZ31 and Ti6Al4V, leading to corrosion damage forming in the depth of the substrate [55].



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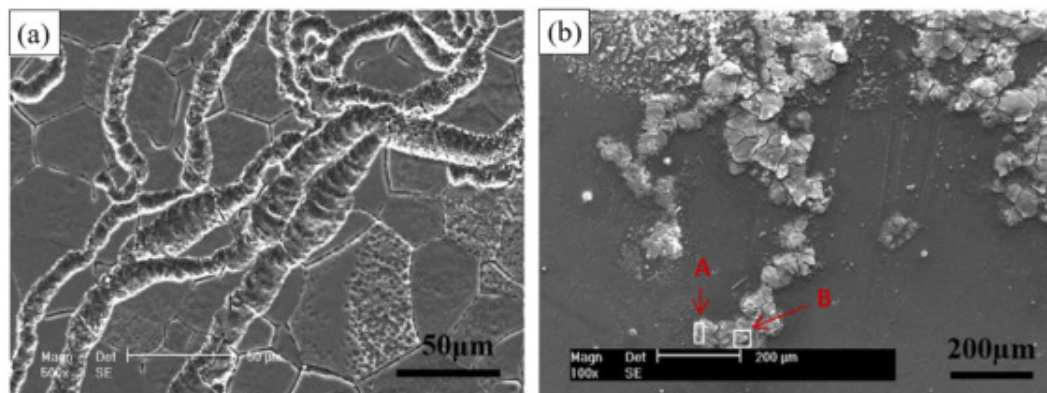
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Fig. 2. Microstructure of galvanic corrosion on AZ31-Ti6Al4V alloy after removal of corrosion products [55].

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Intergranular stress corrosion cracking (ISCC) has been reported during the stress corrosion of magnesium alloys, demonstrating that intergranular corrosion (IGC) can indeed occur in these materials. In particular, the IGC morphology has been observed in as-extruded AZ80 magnesium alloy following T5 aging treatment [56], although the resulting corrosion is relatively shallow. This corrosion pattern arises from the formation of micro-galvanic couples between the  $Mg_{17}Al_{12}$  phase and the surrounding  $\alpha$ -Mg matrix, with the latter undergoing preferential attack.

Filiform corrosion is also one of the most common types of corrosion in magnesium alloys. Unlike steel, where filiform corrosion usually occurs beneath coatings in high-humidity environments, in Mg alloys, it appears under different conditions. It occurs as frequently as pitting corrosion and is typically observed on bare Mg alloy surfaces exposed to air or NaCl solutions. Filiform corrosion begins at corrosion pits and propagates along the active regions of the surface. In magnesium alloys, corrosion predominantly occurs through hydrogen evolution rather than oxygen absorption or oxygen concentration mechanisms. This type of corrosion is driven by active corrosive cells moving across the metal surface. In Mg, filiform corrosion is fueled by the oxygen concentration difference between the filament head and tail, leading to the development of a specific filiform corrosion model for magnesium [57]. Earlier studies suggested that filiform corrosion mainly occurs beneath protective coatings or anodic oxide layers, and that pure, uncoated magnesium does not typically exhibit this behavior [58,59]. A study by [60] and reported by [61] revealed a filiform corrosion morphology of Mg-3 Zn alloy in a NaCl solution, as displayed in Fig. 3. Although, the same alloy sample displayed pitting corrosion when immersed in  $Na_2SO_4$  solution.

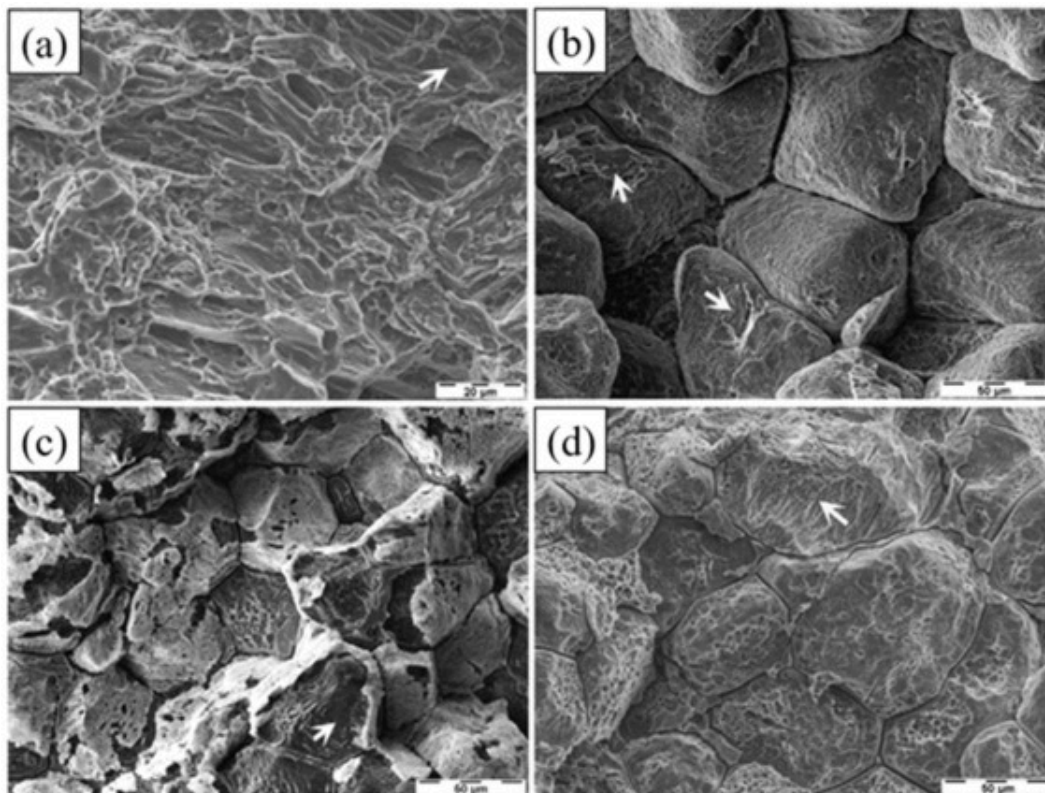


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Fig. 3. Filiform corrosion morphologies of Mg–3 Zn immersed in NaCl solution [60,61].

Stress corrosion cracking (SCC) in magnesium alloys arises from the combined effects of electrochemical reactions and tensile stress. SCC can develop whenever a magnesium-based component under stress is exposed to a wet environment. The primary mechanisms involved are anodic dissolution (AD) and hydrogen embrittlement (HE). Anodic dissolution in Mg alloys is largely due to their highly negative electrochemical potential and the formation of a non-protective corrosion product layer [62,63]. Fig. 4 presents the corrosion micrographs of SCC fractures of some Mg-MMCs and Mg alloys in NaCl solution at a certain strain rate. Stress corrosion is mainly dependent on the material, tensile stress, and environment. SCC resistance is significantly boosted with alloying, while the influence of Al and Zn on the susceptibility of SCC of magnesium is obvious. The environmental influence of SCC is majorly observed in its  $\text{Mg}(\text{OH})_2$  film. Hence, different pH solutions will greatly affect the corrosion behaviour of Mg alloys, leading to a significant effect on the stress corrosion of the Mg alloys [61].



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Fig. 4. Micrograph showing intergranular and transgranular stress corrosion cracking of Mg alloys [61].

Crevice corrosion in metals like iron, stainless steel, and aluminum alloys is typically driven by differences in oxygen concentration (through the oxygen reduction cathodic reaction). In magnesium alloys, however, variations in oxygen concentration are negligible, and as a result, crevice corrosion in Mg was largely overlooked [64]. It was not until 2011 that crevice corrosion was first observed in magnesium, occurring within the crevices of specially treated pure Mg, though the underlying mechanism remained unclear [65]. A comprehensive study on Mg-based composite developed using AZ31 and

Ti6Al4V powders through Pulse Plasma sintering (PPS) was conducted by [55]. Crevice-like corrosion at the matrix-reinforcement interface was addressed by the study. The initiation of the crevice corrosion at the interface is significantly achieved because of the variation in potential difference between the reinforcement (cathodic) and Mg matrix (anodic). There is a corrosion propagation preferentially along the matrix-reinforcement boundary, forming deep channels feature of crevice attack.

### 3. Corrosion behaviour of Mg-MMC

#### 3.1. Acidic media

The corrosion of magnesium metal matrix composites (Mg-MMCs) in acidic environments is a severe and complicated process characterized by the active dissolution of the magnesium matrix, which is greatly impacted by the presence of reinforcing phases. The low pH environment quickly breaks down any magnesium hydroxide ( $\text{Mg}(\text{OH})_2$ ) coating that would provide some protection under neutral circumstances. This prevents the creation of a stable passive layer [66]. Consequently, the corrosion mechanism is essentially electrochemical and chemical in nature, driven by the following underlying reactions, as presented in Eqs. (1) and (2):

The anodic reaction is the oxidation of magnesium:



The predominant cathodic reaction is hydrogen evolution due to the high concentration of  $\text{H}^+$  ions:



The metal is rapidly consumed as a result of this overall reaction, and hydrogen gas evolves rapidly. Furthermore, the microstructural and electrochemical conditions are significantly altered by the addition of reinforcements ( $\text{SiC}$ ,  $\text{Al}_2\text{O}_3$ ,  $\text{B}_4\text{C}$ , and carbon

nanotubes (CNTs)), which introduce two main processes that enhance degradation. The two main processes are micro-galvanic coupling and interfacial degradation, and micro-cracking, and are further discussed.

### 3.1.1. Micro-galvanic coupling

The formation of micro-galvanic cells between the reinforcement and the magnesium matrix is the most critical mechanism. Compared to magnesium, most carbon-based and ceramic reinforcements are more noble and electrochemically inert. Due to this difference, the magnesium matrix is forced to function as a sacrificial anode, which speeds up its disintegration immediately adjacent to the cathodic reinforcement particles [67]. The extent of this acceleration is governed by the cathode-to-anode area ratio, the electrical conductivity of the reinforcement, and its distribution within the matrix. Additionally, the rate of corrosion can be significantly increased by the formation of an effective, broad cathode from highly conductive, continuous networks of reinforcements (e.g., graphene nanoplatelets) [26].

### 3.1.2. Interfacial degradation and micro-cracking

Corrosion preferentially starts at the contact between the reinforcement and matrix, and inherent weaknesses in interfacial bonding, micro-voids formed during processing, or the formation of brittle intermetallic phases can create micro-crevices. These imperfections are easily penetrated by the acidic electrolyte, which causes particle undermining and crevice corrosion. Moreover, residual thermal stresses resulting from the matrix and reinforcement's different coefficients of thermal expansion may cause microcracks, which would facilitate the acid's swift penetration into the bulk material and hasten the corrosion process [68].

In acidic environments, reinforcements that accelerate corrosion involve those with high electrochemical activity or those that form efficient galvanic couples with the magnesium matrix. These "high-acceleration reinforcements" act as effective cathodes, creating a substantial electrochemical potential difference with the anodic magnesium matrix [44].

For instance, metallic particles such as Cu, Ni, and Fe possess high exchange current densities for hydrogen evolution; their low hydrogen overvoltage promotes severe galvanic corrosion [69]. Similarly, carbon-based reinforcements, including carbon fibers, carbon nanotubes (CNTs), and graphene, create powerful galvanic cells with magnesium. In acidic media, hydrogen ions are reduced on the carbon surface, leading to deep pitting at the interface. Furthermore, free carbon present in large-scale boron carbide ( $B_4C$ ) can promote galvanic attack, acting as a focal point for localized corrosion.

While no reinforcement can completely halt corrosion in acidic media, "less detrimental reinforcements" can minimize the acceleration due to their non-conductive nature or lower potential difference relative to the matrix [70]. Ceramic oxides such as  $Al_2O_3$ ,  $ZrO_2$ , and  $TiO_2$  are electrical insulators that cannot act as cathodes for hydrogen evolution, thereby preventing the establishment of strong galvanic cells [61]. Additionally, rare earth oxides like  $Y_2O_3$  can act as corrosion inhibitors; they migrate during the initial stages of attack and incorporate into the surface film, slowing the ingress of acidic ions and slightly stabilizing the composite.

The observed corrosion patterns are more strongly influenced by the interfacial defects of the reinforcements, which can also be linked to the chemical composition of the reinforcements. The more reinforcement in the magnesium alloy, the more the agglomeration of the reinforcements in the magnesium matrix, which acts as an initiation point for corrosion. According to [71], the secondary phases in Mg alloys, i.e., intermetallic phases, trigger micro-galvanic reactions by becoming cathodic sites during their corrosion testing in aqueous solutions.

In acidic environments, the corrosion behavior of Mg-MMCs is significantly influenced by the pH value and the presence of chloride ions ( $Cl^-$ ). Lower pH values accelerate corrosion, while certain reinforcements, such as aluminum-based particles, can mitigate this effect by promoting the formation of more stable mixed-oxide layers [61]. Under acidic conditions, the corrosion behaviour of Mg-MMCs shifts from "passive-active" competition to rapid, continuous dissolution as the protective  $Mg(OH)_2$  film becomes

thermodynamically unstable. Studies have shown that corrosion rates increase exponentially as the pH drops; for example, a shift from pH 7 to pH 5 can increase the corrosion rate of Mg-MMCs by an order of magnitude due to the aggressive nature of hydrogen ions [44]. Furthermore, outcomes for the same Mg-MMCs can vary significantly depending on testing methodologies, such as potentiodynamic polarization versus mass loss. Long-term immersion tests in acidic media tend to reveal that nano-sized particles perform better than micro-sized ones because they provide a more uniform distribution and create smaller, more widely distributed galvanic cells, which prevent the severe localized pitting typical of micro-composites.

### 3.2. Alkaline media

The corrosion behaviour of magnesium metal matrix composites (Mg-MMCs) in alkaline environments is essentially different from that in acidic or neutral media, which are characterized by a substantially lower corrosion rate due to the formation and stabilization of a passive surface film. The high pH environment (usually >10.5) encourages the quick development of a magnesium hydroxide ( $\text{Mg}(\text{OH})_2$ ) layer on the magnesium matrix's surface. This  $\text{Mg}(\text{OH})_2$  film acts as a physical barrier, inhibiting the direct contact between the metal and the corrosive electrolyte and drastically slowing down both anodic dissolution and cathodic reactions as expressed in Eq. (3) [72].



The primary cathodic reaction in deaerated alkaline solutions is the reduction of water (Eq. (4))



The primary determinants of Mg-MMCs' corrosion performance in alkaline conditions are the passive film's stability and protective qualities [61]. Reinforcement particles can enhance this stability if they contain elements like Silicon or Aluminium (e.g., SiC or  $\text{Al}_2\text{O}_3$ ), which react with the magnesium matrix to form a compact, mixed-oxide layer

such as the  $\text{MgAl}_2\text{O}_4$  spinel [73]. This layer is significantly less porous than standard magnesium hydroxide. Furthermore, these particles often act as grain refiners during manufacturing; a finer grain structure increases grain boundary density, promoting a more rapid and continuous formation of the passive film across the surface.

In alkaline media, the reinforcement's function shifts from being a galvanic catalyst to a major factor influencing film structure. However, reinforcement particles still act as initiation sites for localized corrosion if a galvanic mismatch exists. Interface defects, such as porosity gaps, allow alkaline electrolytes to penetrate deeper into the material, initiating internal corrosion. Additionally, the disparity in the thermal coefficient of expansion between the Mg matrix and reinforcements creates high-stress zones during cooling, which are chemically reactive and prone to stress-corrosion cracking. The introduction of chloride ions into alkaline media leads to preferential attack at the interface, where passive film discontinuity makes the particle boundary the path of least resistance for ion penetration [61].

The corrosion mechanism is primarily governed by the following factors:

### 3.2.1. Reinforcement-induced film stability

Stable, inert ceramic reinforcements, such as  $\text{SiO}_2$ ,  $\text{B}_4\text{C}$ , and  $\text{Al}_2\text{O}_3$ , can be included in the expanding  $\text{Mg}(\text{OH})_2$  layer. This can improve the corrosion resistance by increasing the passive film's mechanical stability and density, which will lessen its vulnerability to breakdown and cracking. The particles can serve as tangible anchors, forming a composite film that is more resilient and adhesive [74].

### 3.2.2. Interfacial integrity and localized attack

Localized corrosion initiation may still occur at the reinforcement-matrix interface even while the overall corrosion is controlled. The high pH electrolyte can overcome the weak interfacial connection and pass through the interface. However, in alkaline conditions, the hydrolysis of  $\text{Mg}^{2+}$  ions ( $\text{Mg}^{2+} + 2\text{H}_2\text{O} \rightarrow \text{Mg}(\text{OH})_2 + 2\text{H}^+$ ) can cause the subsequent crevice

chemistry to become acidic, resulting in a regional pH reduction. This acidification inside the crevice can destabilize the passive film locally, leading to minor, confined pitting or crevice corrosion around particles. This is often self-limiting if the film is otherwise stable, as the bulk high pH environment quickly re-passivates the exposed metal [75].

Fluctuating pH levels pose a significant challenge to Mg-MMCs because magnesium's passivity is highly sensitive to the acidity or alkalinity of the local environment [61]. When the pH is not constant, the "steady state" required for a stable film is never truly reached. This prolonged exposure to shifting pH levels leads to the passive film undergoing a cycle of dissolution and reprecipitation, which often leads to structural failure of the protective layer.

The repeated cycles lead to decohesion at the reinforcement-matrix interface, resulting in the particle "falling out" of the matrix, a phenomenon exacerbated by the high electrochemical activity at these junctions. Furthermore, pH fluctuations act as a catalyst for pit growth; a pit initiated during a low pH excursion rarely heals completely when the pH returns to normal, leading to deep, localized penetration [76]. The extent to which corrosion behavior in alkaline media reflects the actual service environments of Mg-MMCs depends on the specific industry. There is a high correlation when Mg-MMCs are used as reinforcement in concrete or as embedded sensors due to the stable, high alkalinity of cementitious pores. Conversely, there are low and variable correlations in marine and industrial environments, and biomedical implants, respectively, where chloride-induced breakdown and fluctuating physiological pH dominate [61]. Magnesium and its alloys rely on the formation of a  $\text{Mg}(\text{OH})_2$  film for protection. According to the Pourbaix diagram for magnesium, this film is only thermodynamically stable in highly alkaline environments ( $\text{pH} > 10.5$ ).

### 3.3. Saline/Chloride-rich media

Magnesium metal matrix composites (Mg-MMCs) are particularly susceptible to corrosion in environments with high salinity or chloride content, as these conditions are

extremely harsh and replicate many real-world applications. Severe localized corrosion results from the chloride ion ( $\text{Cl}^-$ ), which is the main aggressor that quickly breaks up the partially protective magnesium hydroxide ( $\text{Mg}(\text{OH})_2$ ) coating that forms on the surface. Microstructural degeneration, galvanic acceleration, and film breakdown interact intricately in this system [77].

The initial anodic and cathodic reactions are presented in Eqs. (5) and (6), respectively:

Anodic:



Cathodic (in neutral aerated solutions):



The presence of  $\text{Cl}^-$  ions initiates film breakdown through competitive adsorption and formation of soluble magnesium chloride complexes (Eq. (7)):



By destabilizing the passive coating, this reaction exposes the underlying active metal to the electrolyte and hinders efficient repassivation. The following processes significantly deteriorate this process for Mg-MMCs:

### 3.3.1. Intensified galvanic corrosion

In chloride media, this is the predominant mechanism. For the oxygen reduction process (ORP), the noble and conductive reinforcements (e.g., SiC, TiC, CNTs, and graphene) serve as extremely effective cathodic sites. By dissolving the protective layer, the chloride ions guarantee that the cathodic reaction on the reinforcement is always supported by a sizable and actively dissolving anodic region (the magnesium matrix). This produces a strong, long-lasting micro-galvanic coupling, which accelerates and deepens the localized attack at the matrix-particle edge [78]. The ratio of cathode to anode area is crucial

because a larger percentage of contiguous cathodic reinforcement causes the rate of corrosion to rise sharply.

### 3.3.2. Enhanced pitting and crevice corrosion

Mg-MMCs' microstructural diversity offers a large number of pitting starting locations. There is an inherent vulnerability in the reinforcement-matrix interaction. These interfaces, microcracks, and any processing-induced flaws (such as porosity or microcracks) are all aggressively penetrated by chloride ions [79].

### 3.3.3. Filiform corrosion

Filiform corrosion may be the result of chloride-induced corrosion in some circumstances, especially when dealing with particular metal matrices. This happens when a corrosive front that is propelled by differential aeration and chloride concentration moves beneath the delicate  $\text{Mg}(\text{OH})_2$  coating, leaving behind a visible corrosion product trail that resembles a thread. The existence of reinforcements can either function as barriers to the spread of filaments or disturb the continuity of the film, depending on their size and distribution [80].

When considering chloride-induced corrosion in environments like seawater, research indicates that both the matrix composition and reinforcement phase play critical roles [81]. The electrical conductivity and electrochemical nature of the reinforcement, such as carbon fibers or silicon carbide, dictate the rate and severity of the corrosion attack by establishing micro-galvanic cells at the interfaces [44,61]. Conversely, the matrix composition, including the presence of alloying elements like aluminum or rare earths, determines the base susceptibility and the stability of any forming passive film [44,76]. This implies that while the threshold of corrosion initiation depends on the matrix's purity and alloying, the subsequent acceleration is driven by the reinforcement [44].

An electrically conductive and noble reinforcement acts as a "highway" for electron transfer, facilitating cathodic reactions (like hydrogen evolution) that lead to rapid,

localized failure [26]. Even if a corrosion-resistant magnesium alloy is used as a matrix, the addition of highly conductive reinforcements almost always results in a dominant galvanic effect in saltwater [44].

For most magnesium alloys and their composites, there is no specific threshold for significant reinforcement-accelerated corrosion in chloride concentrations. However, below 0.01 M, corrosion is often controlled by the matrix's self-dissolution rate [61,76]. Furthermore, at concentrations above 0.1 M, chloride ions effectively disrupt the protective hydroxide layer, transforming it into soluble magnesium chloride ( $\text{MgCl}_2$ ) [82]. At this stage, the electrical conductivity of the reinforcement becomes the primary "bottleneck" or driver for the reaction rate.

### 3.4. Industrial and biological media

Magnesium metal matrix composites' (Mg-MMCs') corrosion behaviour in biological and industrial settings is extremely complicated and unpredictable because it is controlled by specific chemical species, pH variations, and dynamic circumstances that are absent from typical laboratory solutions. The process frequently combines chemical assault, galvanic effects, and occasionally bio-interactions synergistically [[83], [84]].

#### 3.4.1. Industrial media

Industrial environments can include a wide range of chemicals, but exposure to carbonate/phosphate solutions and engine coolants is particularly relevant for Mg-MMCs. In carbonate/phosphate solutions, surface coatings that are insoluble and extremely protective can develop in solutions that include  $\text{CO}_3^{2-}$  and  $\text{PO}_4^{3-}$  ions, such as in certain cooling fluids or chemical processing streams. More stable and protective than the  $\text{Mg}(\text{OH})_2$  layer in chloride environments, these ions can precipitate with  $\text{Mg}^{2+}$  ions to create compounds such as magnesium carbonate ( $\text{MgCO}_3$ ) and magnesium phosphate ( $\text{Mg}_3(\text{PO}_4)_2$ ).

Active dissolution provides access to a corrosion mechanism governed by the creation and stability of this composite layer. This mixed salt layer can be strengthened by adding inert ceramic reinforcements (such as hydroxyapatite or  $\text{Al}_2\text{O}_3$ ) to increase its density, cohesiveness, and resistance to penetration by chloride ions. When compared to saline medium, this frequently leads to a notable decrease in the corrosion rate [84]. As engine coolant, modern engine coolants are typically alkaline-inhibited ethylene glycol-water mixtures. Their chemistry is designed to be inhibitive towards metals.

The Mg matrix's corrosion is much reduced in these alkaline (pH 8–11) and inhibited conditions. The inhibitors (such as organic acids, nitrites, and silicates) either create protective coatings or adsorb on the metal surface. The reinforcement's main function is microstructural. The galvanic connection between the matrix and reinforcement may be reactivated, resulting in localized attack, if the coolant gets polluted with chlorides or breaks down over time, producing organic acids [77].

### 3.4.2. Biological media

Biodegradable magnesium-MMCs for implant applications (such as those with tricalcium phosphate (TCP) or hydroxyapatite (HA) reinforcements) corrode in a complex biological electrolyte (blood plasma, interstitial fluid) with a pH of around 7.4 to 7.6. The presence of proteins and the way biological deposits build govern the process and make it distinctive [85].

Upon implantation, proteins from blood plasma (albumin, fibrinogen) immediately adsorb onto the composite surface, forming a proteinaceous layer. This is achieved by restricting ion transport. This protein layer can serve as a barrier, initially slowing down corrosion. It also serves as a precursor to the development of a calcium phosphate (Ca-P) layer that is more stable. A coating of calcium phosphate or carbonate apatite that resembles bone forms on the implant surface as a result of blood plasma being supersaturated with  $\text{Ca}^{2+}$  and  $\text{PO}_4^{3-}$  ions. The corrosion rate can be considerably slowed

down by this biomineralized layer, which is extremely protective. Bioactive reinforcements, such as HA, can speed up this process.

The corrosion rates of Mg-MMCs measured in simulated body fluids (SBF (in vitro)) compared to in vivo usually revealed discrepancies. SBF is designed to mimic the ionic concentration of human plasma; however, it fails in accounting for the dynamic biological environment, leading to significant discrepancies in corrosion rates. Most often, SBF tests overestimate the degradation rate compared to in vivo results. The corrosion of Mg alloys and their composites in SBF is a complex process. Studies have shown that in vitro corrosion rates can be between 2 and 4 times higher than in vivo rates for the same magnesium alloy [[86], [87], [88]]. In a living organism, the formation of proteinaceous pseudo-capsule and the presence of organic molecules, such as albumin, can physically block the magnesium surface, slowing down ion exchange in a way that static SBF cannot replicate. As reported by [70], the correlation between in vitro and in vivo corrosion rates is strongly dependent on the electrolyte chosen for the laboratory studies. In vivo, the observed possible reasons for lower corrosion rates compared to in vitro are linked to the lower chloride concentration in blood plasma, or the presence of proteins and cells, and so on. In vivo, magnesium surface adsorbs plasma proteins, whereas SBF lacks protein, which leads to chloride ions attacking the  $\text{Mg}(\text{OH})_2$  layer unimpeded. Furthermore, the pH value changes significantly under the conditions of dynamic fluid flow in vivo [89]. SBF tests are mostly carried out statically, that is, the corrosion products and local pH build up around the sample. For a better evaluation of the corrosion behaviour of Mg-MMCs and to attain a quantitative accuracy threshold in vitro, in vivo implantation is essential [88]. It is proposed that during testing, the pH should be maintained at 7.4 through  $\text{CO}_2$  buffering, protein content inclusion to mimic bio-fouling at the interface, and the use of microfluidic cells to simulate the 0.01–0.1 mL/min dynamic flow rates found in human tissue [89].

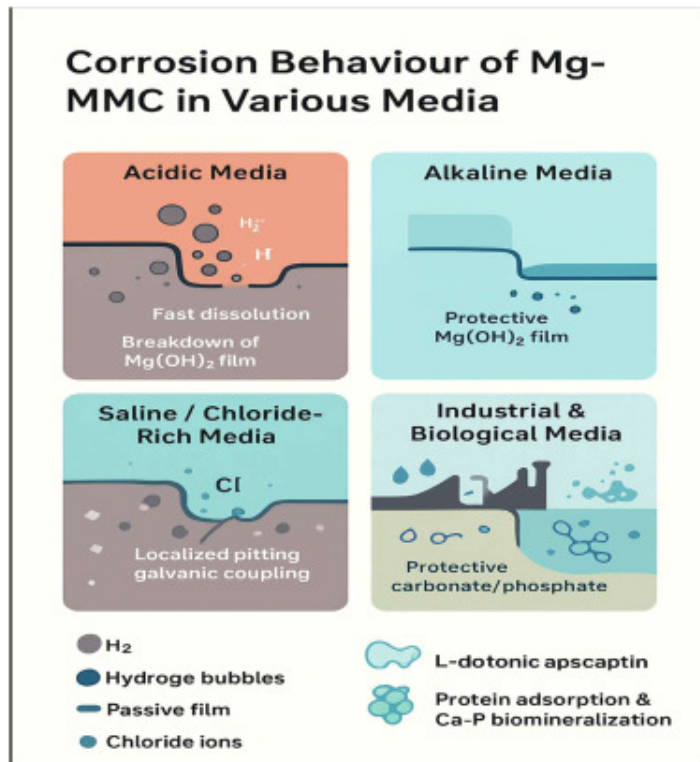
The long-term mechanical stability of Mg-MMC implants is uniquely threatened by the dissolution of the reinforcement phase or the degradation of the interface surrounding it. While the matrix provides the bulk volume, the reinforcement is responsible for the load-

bearing "skeleton" and the overall stiffness (E-modulus) of the implant. The degradation rate of Mg-MMCs allows the mechanical properties of the material to remain intact until the tissue is fully recovered, and a gradual replacement of the implant by regenerative tissue occurs. The Mg-MMCs implant, serving as a biomedical material, must remain stable for 12 weeks [88,89]. Most often, the rate of degradation of these Mg-MMCs is lowered by the incorporation of reinforcement phases. Various factors, including different solutions and immersion time, can significantly influence the corrosion rate of Mg-MMCs. Hence, it is challenging to compare the corrosion resistance of Mg-MMCs in different studies. In a long-term application of Mg-MMCs implants, the ability of the reinforcement to reinforce the matrix alloy is compromised as the corrosion progresses. Prior to the dissolution of the reinforcement itself, the interface usually fails first. In chloride-rich physiological environments, localized galvanic corrosion trenches form around the reinforcement. This results in a stress transfer loss from the matrix to the reinforcement. However, if reinforcement dissolution is too rapid, the implant reaches its critical mechanical failure point before the bone can support the physiological load, leading to secondary fractures. The successful use of Mg-MMCs in biomedical applications requires the dissolution of bioactive reinforcements, including hydroxyapatite,  $\beta$ -Tricalcium Phosphate, or Bioactive Glass, which must be synchronized with both the matrix degradation and the natural bone healing process [88,90]. Hence, the ideal dissolution rate for bioactive reinforcements is generally targeted to be slightly slower than or equal to the degradation rate of the magnesium matrix. It is important to state that excessive bioactivity accelerates localized corrosion because it compromises the structural homogeneity of the material. Under physiological loading, these sites of high bioactivity transform into initiation points for cracks. The implant essentially disintegrates faster than the bone can replace it, leading to a loss of mechanical integrity.

### 3.5. Role of bioactive reinforcements

Reinforcements, like HA and TCP, are not inert; they are soluble and bioactive. The local chemistry may be directly impacted by these reinforcements. As a result of their disintegration,  $\text{Ca}^{2+}$  and  $\text{PO}_4^{3-}$  ions are released, which raises local supersaturation and

speeds up the development of a stable, consistent protective Ca-P layer across the whole composite surface. This lessens the negative micro-galvanic effect that a bio-inert reinforcement would normally have. Thus, surface-mediated biomineralization and regulated, rather uniform deterioration replace the galvanic acceleration-based corrosion mechanism [91]. The corrosion behaviour of Mg-MMC in various corrosion media is displayed in Fig. 5, while Table 2 also highlights the various studies on Mg-MMC's corrosion behaviour.



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Fig. 5. Corrosion behaviour of Mg-MMC in various corrosion media.

Table 2. Various studies on Mg-MMCs corrosion behaviour.

| <b>Mg used</b>           | <b>Reinforcement used</b>             | <b>Corrosion environment</b>                                                                                                                                                                                                                                                                                             | <b>Major findings</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Ref.</b> |
|--------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Magnesium (Mg-Zn matrix) | Nano-sized ZnO particles              | Body fluid (containing NaCl 8.0 g/L, KCl 0.4 g/L, CaCl <sub>2</sub> 0.14 g/L, NaHCO <sub>3</sub> 0.35 g/L, DC <sub>6</sub> H <sub>6</sub> O <sub>6</sub> 0.35 g/L, MgSO <sub>4</sub> ·H <sub>2</sub> O 0.2 g/L, KH <sub>2</sub> PO <sub>4</sub> 0.1 g/L, Na <sub>2</sub> HPO <sub>4</sub> ·12H <sub>2</sub> O 0.06 g/L). | The Mg-MMC demonstrated notable enhancements in mechanical characteristics and corrosion resistance compared to pure Mg. This can be attributed to grain refinement strengthening and the reinforcement of the Mg matrix by the uniform and sequential dispersion of MgO and Mg-Zn intermetallics.                                                                                                                                                                                         | [92]        |
| Magnesium AS41           | Al <sub>2</sub> O <sub>3</sub> fibres | 100 ppm NaCl alkaline solution                                                                                                                                                                                                                                                                                           | Electrochemical test results and free immersion test data showed that the Al <sub>2</sub> O <sub>3</sub> /AS41 Mg MMC demonstrated corrosion behavior similar to that of its pure matrix alloy. Furthermore, there was no apparent change in the corrosion potential or a definite trend in response to the presence of Al <sub>2</sub> O <sub>3</sub> fibres. On the other hand, the presence of Al <sub>2</sub> O <sub>3</sub> was considered to have increased the degree of corrosion. | [93]        |
| Mg <sub>2</sub> Zn alloy | Al <sub>2</sub> O <sub>3</sub>        | Hank's solution (biomedical fluid)                                                                                                                                                                                                                                                                                       | The corrosion resistance of specimens with a high reinforcement ratio is considerably diminished. At the grain boundaries, galvanic corrosion is at its lowest when Al <sub>2</sub> O <sub>3</sub> reinforcement is applied at a modest                                                                                                                                                                                                                                                    | [79]        |

| <b>Mg used</b>                 | <b>Reinforcement used</b>     | <b>Corrosion environment</b> | <b>Major findings</b>                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Ref.</b> |
|--------------------------------|-------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                                |                               |                              | ratio (2 wt%). This facilitates the development of the apatite structure, which enhances the magnesium matrix composite material's resistance to corrosion.                                                                                                                                                                                                                                                                                          |             |
| WZ73 Mg alloy                  | SiC particles                 | 1 wt% NaCl                   | SiC negatively affects the WZ73 alloy's ability to withstand corrosion. The findings of the SiC immersion test show that the MMCs corrode more quickly. Additionally, SiC has an indirect impact on microgalvanic corrosion in the WZ73-SiC MMCs. The Mg matrix/SiC contact at the long period stacking ordered (LPSO) phase disrupts the continuity of the matrix, which speeds up corrosion and results in the higher corrosion rate seen in MMCs. | [94]        |
| AZ31 and AZ61 magnesium alloys | Graphene nanoplatelets (GNPs) | 3.5 wt% NaCl                 | The AZ31 and AZ61 alloy matrices' corrosion currents increased as the GNPs were added. This occurs because dislocation concentrations at the composites' interface produce crevice assaults, which raise the composites' corrosion rate in comparison to unreinforced metals. Preliminary findings indicate that                                                                                                                                     | [95]        |

| Mg used               | Reinforcement used                                                                                        | Corrosion environment | Major findings                                                                                                                                                                                                                                                                   | Ref. |
|-----------------------|-----------------------------------------------------------------------------------------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
|                       |                                                                                                           |                       | the GNPs served as efficient cathodes to accelerate the corrosion.                                                                                                                                                                                                               |      |
| AZ91D magnesium alloy | Nano metal oxides such as Zinc oxide (ZnO), Manganese oxide (MnO), and Titanium oxide (TiO <sub>2</sub> ) | 2.5 wt% NaCl          | According to the electrochemical corrosion data, adding nano oxide reinforcement improves corrosion resistance in comparison to AZD alloy.                                                                                                                                       | [96] |
| AZ91D magnesium alloy | Ti6Al4V                                                                                                   | 3.5 wt% NaCl          | According to the analysis of the potentiodynamic polarization curves and EIS, the corrosion resistance of the composite diminishes with the addition of TC4 particles.                                                                                                           | [97] |
| AZ91D magnesium alloy | Hydroxyapatite (HA) particles                                                                             | Artificial sea water  | Mechanical and corrosive characteristics are significantly influenced by the size and distribution of the HA particles. Contrary to earlier research, adding HA particles improved the magnesium matrix's ability to withstand corrosion in cell fluids and artificial seawater. | [98] |
| AZ91D magnesium alloy | Al <sub>2</sub> O <sub>3</sub> and SiC particles                                                          | 3.5 wt% NaCl          | The result revealed that the surface treatment enhanced the corrosion resistance of the two AZ91D samples, but Al <sub>2</sub> O <sub>3</sub> particles helped in                                                                                                                | [28] |

| <b>Mg used</b>       | <b>Reinforcement used</b>           | <b>Corrosion environment</b> | <b>Major findings</b>                                                                                                                                                                                                                                                                                                          | <b>Ref.</b> |
|----------------------|-------------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                      |                                     |                              | achieving better corrosion resistance compared to SiC particles.                                                                                                                                                                                                                                                               |             |
| Pure Magnesium       | Cu and Mo                           | 3.5 wt% NaCl                 | The corrosion rates of composites were higher compared to pure magnesium. Additionally, the corrosion rate increased with increasing volume fraction of reinforcement in both composites.                                                                                                                                      | [99]        |
| AZ31 magnesium alloy | Tungsten carbide (WC) nanoparticles | 3.5 wt% NaCl                 | The morphology of corrosion indicates the presence of pores and cracks on the corrosion surface, through which $\text{Cl}^-$ ions penetrate and quicken the pace of corrosion. Additionally, more number of cracks at the corroded surface of composites becomes correlated linearly with a reduction in corrosion resistance. | [100]       |
| Pure magnesium       | Graphene Nanoparticle (GNP)         | 3.5 wt% NaCl                 | The concentration of graphene has a detrimental effect on magnesium's corrosion performance. Magnesium shows anodic behaviour while graphene exhibits cathodic behaviour, which leads to micro-galvanic reactions.                                                                                                             | [101]       |

| <b>Mg used</b> | <b>Reinforcement used</b>   | <b>Corrosion environment</b> | <b>Major findings</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Ref.</b> |
|----------------|-----------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Mg-Al alloy    | TiC particles               | 3.5 wt% NaCl                 | The inclusion of TiC particles reduced the base alloy's resistance to pitting corrosion and its uniform corrosion rate, according to all applicable methods. There was some indication of a galvanic reaction between the TiC particles and the matrix. Furthermore, pitting corrosion was a propensity that both the base alloy and the composite exhibited. However, the base alloy's corrosion-controlling factor appeared to be the adsorption of chloride ions, while the composite's rate-controlling element was the diffusion of aggressive ions. | [102]       |
| Mg alloy       | Various rare earth elements | 3.5 wt% NaCl                 | The inclusion of various earth elements, such as Ce, Dy, Gd, Nd, and La, to the Mg alloy individually resulted in a considerable reduction in corrosion mechanism, changing from uniform to localized corrosion mechanism depending on the specific rare earth alloying elements. The formation of the Dy <sub>2</sub> O <sub>3</sub> oxide prevents the Mg-0.41Dy alloy from pitting corrosion and leads to an                                                                                                                                           | [103]       |

| <b>Mg used</b>       | <b>Reinforcement used</b>             | <b>Corrosion environment</b> | <b>Major findings</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Ref.</b> |
|----------------------|---------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                      |                                       |                              | excellent corrosion surface even after 24 h of immersion.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |
| Mg alloy             | Y alloy                               | NaCl                         | The study revealed that corrosion mechanisms of Mg-(0.25 and 2.5) Y alloy and Mg-(5, 8, and 15) Y alloy were uniform corrosion and pitting corrosion, respectively. The content of Mg <sub>24</sub> Y <sub>5</sub> phases determines its effect, acting as a cathode to accelerate the corrosion or as a corrosion barrier to inhibit the corrosion. Y could significantly improve the corrosion resistance of the Mg-Y alloy. However, the corrosion resistance of the Mg-Y alloy deteriorated with the excessive addition of Y into the Mg matrix. | [104]       |
| AZ92 magnesium alloy | silicon-carbide-particle (SiCp)       | 3.5 wt% NaCl                 | AZ92/SiCp composite materials disclosed higher corrosion resistance in high humidity atmosphere (98% RH and 50 °C) with almost no influence of SiCp addition on the corrosion behaviour.                                                                                                                                                                                                                                                                                                                                                             | [105]       |
| Pure magnesium       | Titanium diboride (TiB <sub>2</sub> ) | Simulated body fluid (SBF)   | The incorporation of TiB <sub>2</sub> particles into the magnesium matrix enhanced corrosion-resistant                                                                                                                                                                                                                                                                                                                                                                                                                                               | [106]       |

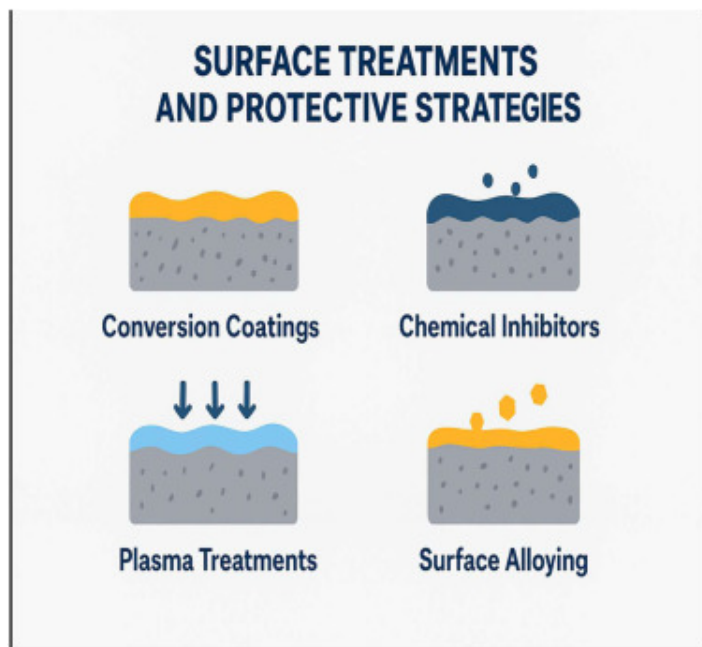
| <b>Mg used</b>       | <b>Reinforcement used</b>             | <b>Corrosion environment</b>                | <b>Major findings</b>                                                                                                                                                                                                                                                                                                                 | <b>Ref.</b> |
|----------------------|---------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                      |                                       |                                             | behavior by improving the physical barrier performance at the metal/solution interface, according to an in vitro corrosion test conducted in SBF medium.                                                                                                                                                                              |             |
| Pure magnesium       | Hydroxyapatite (HA) particles         | Dulbecco's phosphate-buffered saline (DPBS) | The corrosion performance of Mg seems to be affected by competition between grain size and texture. Although Mg-HA composite possesses the finest grain size, the HA particles reduce the basal texture intensity and probably break the continuity of the Mg matrix, which has detrimental effect on the corrosion resistance of Mg. | [107]       |
| Pure magnesium       | Multi-walled carbon nanotubes (MWCNT) | 3.5 wt% NaCl                                | Compared to pure magnesium and grain-refined friction stir-processed magnesium, the corrosion performance of the composites was discovered to be worse because of the inclusion of CNT, which accelerated galvanic corrosion.                                                                                                         | [108]       |
| ZM51 magnesium alloy | SiC                                   | 3.5 wt% NaCl                                | The rough-ground surface displayed the highest corrosion current because of the deep machining scored marks that were associated                                                                                                                                                                                                      | [109]       |

| Mg used | Reinforcement used | Corrosion environment | Major findings                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Ref. |
|---------|--------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
|         |                    |                       | with fractured SiC particles and crevices, while the fine-ground and polished specimens had a relatively low corrosion current density because of their superior surface quality in the potentiodynamic polarization test. The specimen polished with diamond paste had better corrosion resistance than the diamond-turned specimen. The dislodging of SiC particles from the matrix resulted in the formation of fissures at the matrix/particle boundaries. |      |

#### 4. Protective strategies

Magnesium metal matrix composites (Mg-MMCs) tend to corrode quickly, which severely limits their use in demanding industries, including biomedical engineering, automotive, and aerospace. This susceptibility is inherently associated with magnesium's thermodynamic instability and can occasionally become severe in composite systems by the formation of micro-galvanic cells between the cathodic reinforcing phases (SiC, Al<sub>2</sub>O<sub>3</sub>, and B<sub>4</sub>C) and the anodic magnesium matrix [70]. These cells, especially at the matrix-reinforcement interface, speed up localized deterioration. As a result, applying advanced surface engineering techniques is not only advantageous but also necessary to separate the composite from its corrosive surroundings and make use of its remarkable strength and stiffness. The proven and new surface protection techniques designed specifically for

Mg-MMCs are thoroughly examined in this section [110]. The various surface treatments and protective strategies for magnesium metal matrix composites are shown in Fig. 6.



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Fig. 6. Surface treatments and protective strategies for magnesium metal matrix composites.

#### 4.1. Surface modification and conversion coatings

Conversion coatings are created by chemically converting the magnesium-MMC's outermost surface into an inert, highly adherent layer of compounds that acts as a barrier against corrosive substances [111].

##### 4.1.1. Plasma electrolytic oxidation (PEO)

PEO is widely considered the most robust treatment for conductive Mg-MMCs. For light metals, this high-voltage electrochemical method is one of the best surface treatments. Through plasma-driven reactions at the metal/electrolyte interface, a thick, hard, ceramic-like oxide coating is formed, mostly consisting of MgO, MgAl<sub>2</sub>O<sub>4</sub>, and silicate phases. The resulting layer has remarkable thermal stability, wear resistance, and hardness. PEO is highly effective because it can incorporate or cover conductive reinforcements exposed at the surface, breaking the galvanic circuit. Achieving a consistent coating morphology that conformally covers the reinforcing particles and the matrix is a major problem unique to MMCs [112]. Defects may emerge across secondary stages as a result of uneven growth, opening the door for corrosive assault. To lessen this problem and guarantee that the reinforcement is properly contained inside the oxide matrix, process optimization is essential. This includes electrolyte composition (such as silicates, phosphates, hydroxides), as well as electrical regime (current mode, frequency) [113]. PEO layers modified with nanoparticles significantly reduce the corrosion current density of Mg-MMCs through the provision of a physical barrier, which hinders electrolyte penetration to the conductive reinforcements. A review by [114] discussed the growth and corrosion of PEO coating, which established that the overall structure of PEO coatings remains essentially the same, consisting of an internal dense layer and an external porous layer.

## 4.1.2. Chemical conversion coatings

These are cost-effective treatments achieved by immersing the substrate in a specific chemical bath.

### 4.1.2.1. Phosphate-permanganate coatings

These coatings generate a glass-like, amorphous layer consisting of magnesium phosphate and manganese oxides. Their main purpose is to create a better surface for the attachment of subsequent organic paint systems, which significantly increases the protection system's overall longevity, even if they only provide modest corrosion resistance [115].

#### 4.1.2.2. Fluoride conversion coatings

Magnesium fluoride ( $\text{MgF}_2$ ) forms a thin, compact, and extremely insoluble coating after being treated with hydrofluoric acid (HF) or fluoride salts. This coating has been demonstrated to improve biocompatibility and decrease hydrogen gas development in vivo, making it especially beneficial for biomedical magnesium MMCs. It also considerably slows corrosion in physiological fluids that are high in chloride [116].

### 4.2. Applied protective coating systems

This category encompasses techniques that deposit a distinct, functional layer onto the prepared Mg-MMC surface. Fig. 6 displays typical ways of protecting corrosion on the Mg matrix alloy.

#### 4.2.1. Organic/polymer coatings

These coatings, which include acrylics, polyurethanes, and epoxies, serve mainly as high-performance barrier layers that physically isolate the substrate from oxygen and moisture. Surface pre-treatment and adherence to the MMC's frequently uneven and inert surface are crucial to its effectiveness [117]. Creating them as smart coatings can significantly increase their defensive potential. This entails adding nanocontainers that are already laden with corrosion inhibitors, such as cerium ions or green inhibitors like tannic acid, halloysite nanotubes, or mesoporous silica nanoparticles. In reaction to local pH shifts induced by the initial stages of corrosion, these containers offer regulated, on-demand release of the inhibitor, providing active healing activity [118].

#### 4.2.2. Sol-gel derived coatings

The foundation of sol-gel technology is the hydrolysis and condensation of metal alkoxide precursors, such as zirconium propoxide and tetraethyl orthosilicate, to create a dense, inorganic-organic hybrid network at low temperatures. Dip, spin, or spray-coating are methods of application for these coatings, which have outstanding adherence to metallic surfaces.

One significant achievement is the creation of nanocomposite sol-gel coatings, in which functional nanoparticles (clay nanotubes, SiO<sub>2</sub>, TiO<sub>2</sub>, and CeO<sub>2</sub>) are distributed throughout the sol-gel matrix. These nanoparticles serve as fillers, significantly enhancing the mechanical integrity of the coating and the tortuosity of the diffusion channel for corrosive species, which produces a better barrier effect [119].

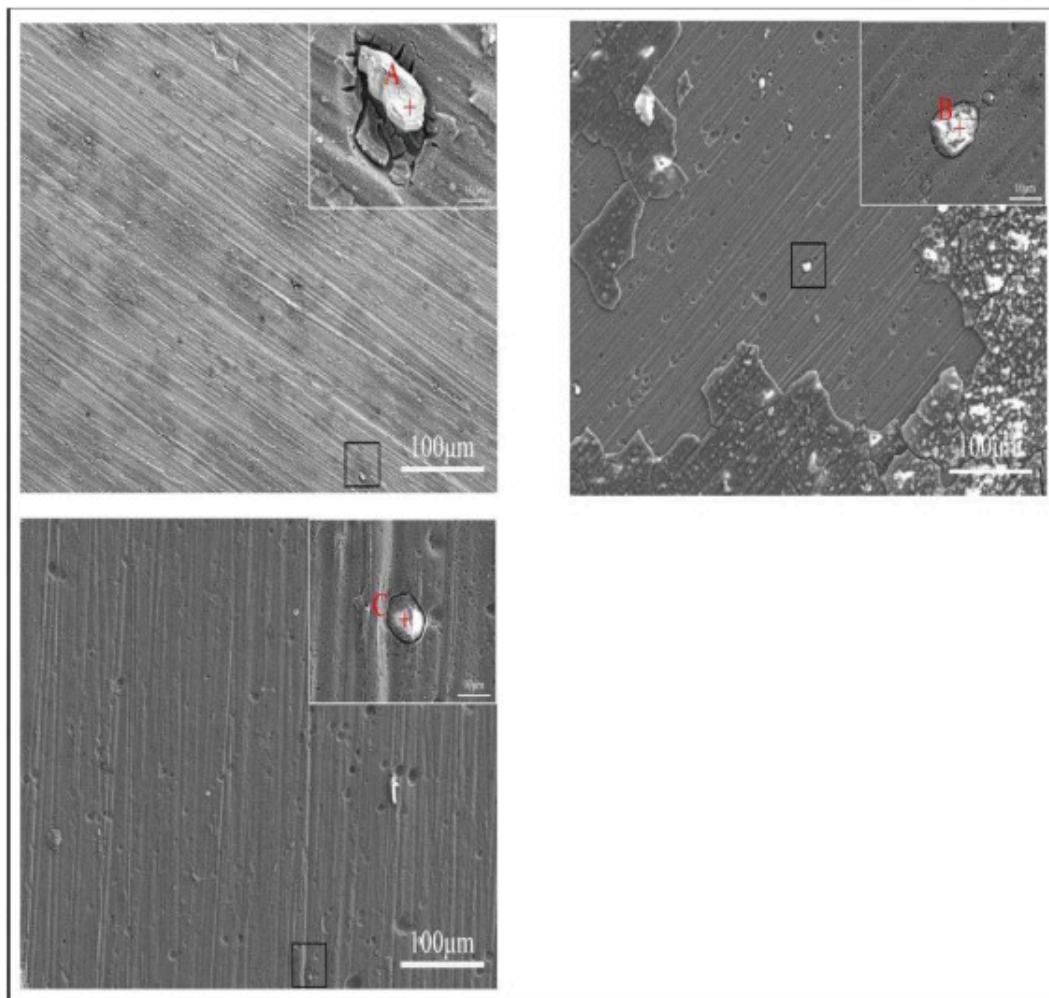
The main failure mechanisms of coatings, particularly on Mg-MMC substrates, can be categorized into mechanical, electrochemical, and interface-specific drivers. These failure mechanisms include cathodic delamination or micro-galvanic failure, coefficient of thermal expansion (CTE) mismatch and thermal spallation, and interfacial debonding or weak boundary layers [120,121]. For the micro-galvanic failure, if the coating is breached, there is formation of galvanic cells where conductive reinforcements act as local cathodes. At the reinforcement site, the reduction reaction yields hydroxide ions (OH<sup>-</sup>), creating a highly alkaline environment at the interface [122,123]. Hence, this alkalinity attacks the chemical bonds of organic coatings or dissolves the oxide layer of the Mg matrix. This phenomenon leads to the peeling of the coating from the reinforcement sites [124,125].

Furthermore, the magnesium matrix expands significantly under thermal cycling more than ceramic reinforcements and ceramic coating. This leads to the generation of massive internal shear stresses due to the CTE mismatch [126]. The coating may undergo flaking off (spallation) or develop a network of mud-cracking to relieve the stress. If the coating process involves aggressive chemicals, the chemicals can seep into the micro-gaps between the reinforcement and the magnesium matrix. This results in internal corrosion underneath the coating before it even fails externally. Due to this, blistering occurs, in which the coating bulges outward, due to the accumulation of hydrogen gas (H<sub>2</sub>) produced by the corrosion of magnesium [127].

## 5. Microstructural behaviour of Mg-MMCs in corrosive environments

The corrosion resistance of Mg-MMCs is determined by the complex interactions between the primary magnesium matrix, the reinforcing phase [128], secondary intermetallic compounds, and the interfaces that bind these constituents together [129]. Micro-galvanic coupling between these features typically initiates pitting, intergranular corrosion, and in some cases, filiform attack.

The microstructural behaviour of Mg-MMCs in corrosive media has been well documented through Scanning Electron Microscopy (SEM) studies. The incorporation of titanium-based reinforcements in magnesium matrix composites has been shown to significantly influence microstructural evolution [130]. In AZ31 Mg alloy reinforced with Ti6Al4V via pulse plasma sintering, refinement of grains and improved densification were observed, leading to enhanced microhardness [131]. Yu et al. [132] demonstrated that Al-rich intermetallic, such as Al-Mg, promotes pitting corrosion in 3.5% NaCl Solution. SEM micrographs from the study confirmed that corrosion preferentially initiates at intermetallic-rich regions and gradually spreads into the matrix, as shown in Fig. 7.



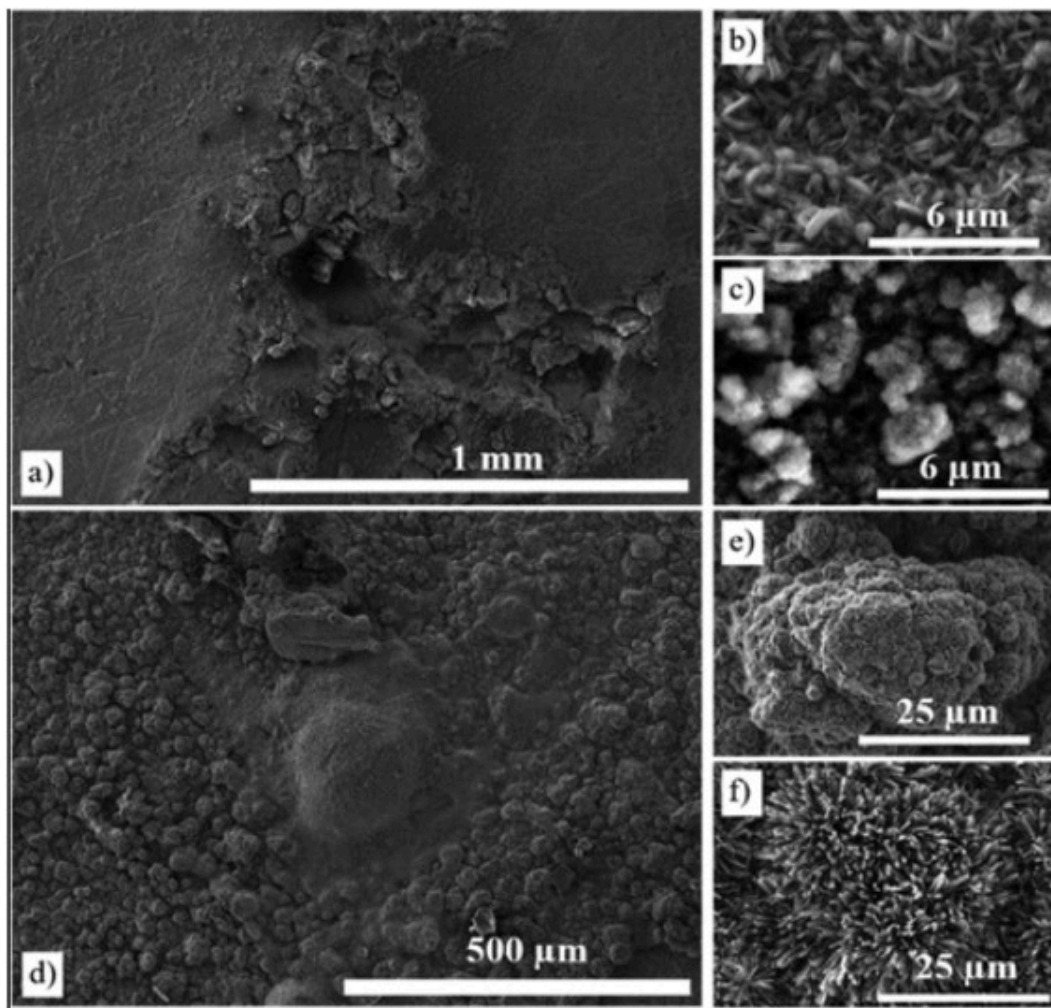
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Fig. 7. SEM surface morphologies of the Al-Mg after the immersion in 3.5 wt.% NaCl [61].

[133] evaluated the corrosion performance of AZ31 Mg alloy in physiological saline and 3.5% NaCl solutions. After 14 days of immersion, the alloy surface in both environments was fully covered by corrosion products (Fig. 8). In the saline solution, residual manufacturing scratches remained visible (Fig. 8a), whereas in 3.5% NaCl solution, a

thicker corrosion layer formed (Fig. 8d). Microstructural analysis revealed heterogeneous corrosion morphologies, including leaf-like structures (Fig. 8b), globular formations (Fig. 8c–e), and needle-like clusters (Fig. 8f), with the latter oriented perpendicular to the surface within the 3.5% NaCl medium. Table 3



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Fig. 8. Surface SEM images of the corroded uncoated AZ31 Mg alloy in saline (a–c) and 3.5% NaCl solutions (d–f) after 14 days [133].

Table 3. Advantages and disadvantages of some surface modification and conversion coating techniques.

| S/N | Surface modification and Conversion coatings | Advantages                                                                                                                                                                   | Disadvantages                                                                                                                                                                                |
|-----|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | PEO/Anodic Oxidation                         | Produces hard, wear-resistant, thermally stable ceramic-like coatings with good adhesion (formed in-situ) [134]                                                              | Porous/coarse morphology can give pathways for corrosion unless sealed or post-treated.                                                                                                      |
| 2   | Phosphate-Permanganate Conversion Coatings   | Improved corrosion resistance relative to untreated substrate. The manganese oxide or phosphates act as a passive barrier [135].                                             | Phosphate-permanganate coatings are often relatively thin and may have porosity, micro-defects. Protection is not always sufficient in harsh environments without additional topcoats [136]. |
| 3   | Sol-gel coatings                             | Highly tunable chemistry (barrier, hydrophobic, corrosion-inhibiting, multifunctional), potential for low-temperature application [137].                                     | Defect control (cracks, pinholes) and long-term durability in harsh environments can be a challenge; often used as topcoats/primers rather than stand-alone heavy-duty protection [138].     |
| 4   | Fluoride Conversion Coatings (FCC)           | Good initial corrosion resistance, especially for magnesium alloys in physiological or chloride-rich environments. It reduces corrosion current density significantly [138]. | Defects (pores, micro-cracks) due to hydrogen evolution during formation, internal stresses, or non-uniform deposition. These act as paths for corrosive species [139].                      |

| S/N | Surface modification and Conversion coatings | Advantages                                                                                                                                                                | Disadvantages                                                                                                                   |
|-----|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 5   | Organic/Polymer Coatings                     | Some polymer systems perform well under UV, humidity, temperature variation, and salt spray. Multi-layer systems (primer + topcoat) often give longer service life [140]. | Pinholes, microcracks, and pores can occur during curing, application, or over time, which let corrosive species through [141]. |

## 6. Future perspectives

The future of magnesium metal matrix composites (Mg-MMCs) lies in the development of corrosion-resistant systems tailored for advanced and emerging applications. The exploration of nanostructured reinforcements and hybrid composites offers significant potential for enhancing both mechanical and electrochemical performance. Mg-MMCs intended for marine environments, offshore structures, and biomedical implants require adaptive protection methods capable of resisting localized attack under highly variable conditions. Furthermore, the application of machine learning and computational modeling to predict corrosion behavior could accelerate the design of next-generation Mg-MMCs. Sustainability should also guide future research, with emphasis on recyclable reinforcements, green corrosion inhibitors, and eco-friendly coating technologies. By bridging experimental insights with predictive simulations and industry-driven requirements, Mg-MMCs can be engineered for reliable use in critical sectors such as aerospace, defense, marine engineering, and healthcare.

It is generally accepted that Mg-MMCs offer an incredible strength-to-weight ratio and have other advantages that make them useful. However, the industrial application of Mg-

MMCs faces some primary research challenges, which are discussed and would require further studies in these areas

- 1 Interfacial incompatibility and wettability: Magnesium has poor wettability with most synthetic/ceramic reinforcements; that is, during casting, the molten metal does not stick to the ceramic particles. This phenomenon leads to agglomeration (clumping) and a weak interface. Consequently, premature brittle failure and low ductility are achieved owing to the material developing micro-cracks at the weak boundaries since loads are transferred from the strong reinforcement to the weak areas. Cost-effective surface treatments or coupling agents could bridge the chemical gap between the organic/ceramic reinforcement and the matrix alloy.
- 2 Micro-galvanic corrosion acceleration: Most reinforcements used in Mg-MMCs development are more noble than the matrix alloy. The exposure of the developed composite to moisture or an aggressive environment leads to the reinforcement particle acting as a tiny cathode, while the magnesium matrix acts as the anode. Consequently, several micro-galvanic cells are initiated across the surface. This leads to localized pitting and tunnelling corrosion, which degrades the structural integrity much faster than the unreinforced matrix alloys.
- 3 Industrial scalability: Magnesium is highly reactive and prone to oxidation during stirring. In large-scale stir casting, it is extremely difficult to keep nanoparticles uniformly suspended. However, they either float to the top or sink to the bottom due to density differences. Hence, there is a heterogeneous distribution of the reinforcements in an industrial casting; that is, some parts have rich and lean areas of reinforcement. This implies that the part's performance is unpredictable, which is unacceptable for safety-critical industries like aerospace or automotive.

Further research should utilize advanced additive manufacturing (3D Printing) or ultrasonic-assisted casting as the production technique rather than the traditional casting methods.

The application of machine learning (ML) for corrosion prediction is feasible; however, careful consideration of the data constraints is required. Big data in corrosion science is a luxury. Most experimental datasets contain fewer than 100 samples because corrosion tests are time-consuming and expensive. The application of specific machine learning strategies is highly effective for predicting corrosion rates, pit depths, and inhibitor efficiency. Rather than using raw data, domain-specific descriptors, such as pH, chloride concentration, or electrochemical potential, can be used to guide the model. Models like Random Forest, Support Vector Machines, and Bayesian Ridge Regression are robust against overfitting in small datasets compared to deep learning. Moreover, the hybrid approach (Physics-Informed Machine Learning (PIML)) can be adopted. In this approach, the use of physical laws guides the development of the ML model. The model requires significantly less data because the search space for a solution is narrowed by known science. Hence, for future study, ML should be integrated with electrochemical principles with data-driven learning for modeling galvanic coupling between matrix and reinforcement. Also, there is limited exploration of microstructural defect detection, phase identification, and reinforcement dispersion analysis using ML. Hence, computer vision integration, such as Convolutional Neural Networks (CNNs) and vision transformers, can be applied to scanning electron microscopy (SEM) or transmission electron microscopy (TEM) images to quantify porosity level, reinforcement agglomeration, and grain boundary characteristics of the magnesium matrix composites.

The sustainable measures to be prioritized in future research on Mg-MMCs corrosion are highlighted, including:

- 1 Future research must move beyond laboratory corrosion rates and look at the life cycle assessment (LCA) of corrosion management, that is, cradle-to-grave impact.

- 2 Research should prioritize Mg-MMCs that are easily recyclable. Many high-value magnesium alloys are turned into low-grade scrap because of the ceramic reinforcements, such as SiC or Al<sub>2</sub>O<sub>3</sub>, which are difficult to separate from the matrix.
- 3 Future research should prioritize the utilization of more traditional bio-based and green corrosion inhibitors. Many more plant-derived extracts should be explored as surface treatments or matrix additions. Also, more self-healing coatings could be explored.
- 4 More agricultural or industrial waste as reinforcement in the development of Mg-MMCs can be explored. Rather than using energy-intensive synthetic ceramics, sustainability is achieved by using these waste materials as reinforcements.

## 7. Conclusion

Magnesium metal matrix composites (Mg-MMCs) exhibit outstanding potential for structural and biomedical applications due to their high strength-to-weight ratio, biocompatibility, and multifunctional properties. However, their long-term performance is critically challenged by corrosion, which is strongly influenced by matrix–reinforcement interactions and environmental conditions. This review highlighted the mechanisms of Mg-MMC degradation in acidic, alkaline, saline, industrial, and biological media, while emphasizing surface treatments and protective coatings as promising solutions. Advancing corrosion-resistant Mg-MMCs through tailored reinforcement strategies and innovative protection systems is essential to fully unlock their potential in demanding engineering and biomedical applications.

## CRediT authorship contribution statement

**Peter P. Ikubanni:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Formal analysis, Data curation, Conceptualization. **Otaigbe C. Anetor:** Writing – review & editing, Writing – original draft, Validation, Formal analysis. **Daniel C. Okolo:** Writing – review & editing, Writing – original draft, Validation, Formal analysis, Data curation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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